

Nicholas Hirsch

nicholashirsch@ufl.edu | 352-359-7371 | nicholashirsch.github.io | linkedin.com/in/nicholas-hirsch | US Citizen

Education

University of Florida, B.S. in Aerospace Engineering (minoring in Computer Science)

December 2027

- GPA: 3.90/4.00
- Benacquisto Scholar (full-ride scholarship)

Experience and Extracurriculars

Starlink GNC Intern, SpaceX – Redmond, WA

August 2026 – Present

- Worked on constellation design and optimization as a member of the Starlink fleet operations team

Astrodynamics Intern, Saber Astronautics – Boulder, CO

May 2026 – August 2026

- Led full-stack development for a conjunction assessment (CA) engine deployed on Saber's space-battle management software (SBMS) whose results are compliant with Space Force unified data library schema
- Wrote highly-optimized C++ CA prescreening and probability of collision algorithms (accounting for complexity in both time and space) designed to regularly screen all commercially-available ephemeris
- Setup containerization (via Docker) to Saber's Kubernetes stack, automatic upload/download from Amazon-hosted SQL databases, and RESTful API access hooks (using OpenAPI schema)

Lead Developer, HohmannPy – Gainesville, FL

Dec 2025 – Present

- Developed an orbital mechanics library and visualization package in Python (<https://github.com/hohmannpy/hohmannpy>) for use in rapid spacecraft design and iteration
- Implemented Kepler, universal variable, Cowell, and Encke method orbital propagators capable of parallel processing large satellite constellations (tested with 10k RSOs) as well as quaternion-based attitude dynamics
- Programmed user-friendly orbit determination through both classical and modified equinoctial orbital elements as well as via Gibbs' method and a Lambert problem solver
- Modeled geopotential, solar radiation, N-body, and atmospheric drag special perturbations as well as impulsive and continuous burn scheduling

Vice President of Flight Vehicles, Florida Rocket Lab – Gainesville, FL

Oct 2024 – Present

- Directed avionics, structures, and GNC teams in designing a UF's first liquid-propelled rocket, Mirage, for the FAR OUT competition; achieved successful launch and recovery
- Wrote custom trajectory simulators in MATLAB and Python for apogee prediction worth 50% of competition scoring; predicted true apogee with only 6% error
- Performed CFD as well as nonlinear optimization to size the fins, ensuring static stability for the rocket
- Developed and led an 80 person new member project over the course of two months that taught basic controls concepts
- Designed an LQI controller in MATLAB/Simulink and C++ for a mono-propelled VTOL drone
- Conducted a robustness analysis to determine the stability of the controller under both crosswinds and Gaussian noise
- Deployed a Kubernetes cluster using ArgoCD to host containerized versions of team simulation code

Researcher, Vehicle Dynamics and Optimization Lab – Dr. Anil V. Rao – Gainesville, FL

Nov 2023 – Aug 2024

- Built a transcription optimal control software (<https://github.com/nicholashirsch/universal-optimal-control-software>) in MATLAB which is problem-agnostic and used it to control a variety of nonlinear systems such as a 6-DOF robotic arm, simplified 2-DOF moon-lander, and the Brachistochrone problem
- Designed and implemented fourteen additional small optimal controllers using both direct and indirect methods

Researcher, Smart Autonomy Lab – Dr. Yu Wang – Gainesville, FL

Jun 2024 – Present

- Deployed state-machine augmented deep reinforcement learning agents across OpenAI Gymnasium environments in Python to test the ability of agents to solve multi-step gridworld tasks
- Implemented a proximal policy optimization algorithm using MLPs via PyTorch to solve cart-pole problems
- Co-authored and submitted a hybrid deep reinforcement learning and supervisory control framework (ICML 2025)
- Served as a teaching assistant for an undergraduate linear control course

Skills and Coursework

Skills: MATLAB/Simulink, Python, C++, C#, JavaScript/node.js, Java, SQL, Git, Agile Development, STK, Linux, Kubernetes, RESTful APIs, Docker, Ansys Fluent, Parallel Computing

Coursework: Dynamics, Vibrations, Linear Controls, Astrodynamics, Flight Dynamics, Spacecraft Dynamics, Numerical Methods, Kalman Filtering, Discrete Mathematics